

1 Steady Conduction

Constants

Stefan-Boltzmann constant

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$$

Formulas

Conduction resistance through a wall

$$R_{\text{wall}} = \frac{L}{kA} \quad (1.1)$$

Conduction resistance through a cylinder

$$R_{\text{cyl}} = \frac{\ln(r_2/r_1)}{2\pi Lk} \quad (1.2)$$

Conduction resistance through a sphere

$$R_{\text{sph}} = \frac{r_2 - r_1}{4\pi r_1 r_2 k} \quad (1.3)$$

Convection resistance

$$R_{\text{conv}} = \frac{1}{h_{\text{conv}}A} \quad (1.4)$$

Radiation resistance

$$R_{\text{rad}} = \frac{1}{h_{\text{rad}}A} \quad (1.5)$$

Radiation heat transfer coefficient

$$h_{\text{rad}} = \varepsilon\sigma(T_1^2 + T_2^2)(T_1 + T_2) \quad (1.6)$$

Radiation heat transfer

$$\dot{Q}_{\text{rad}} = \varepsilon\sigma A(T_1^4 - T_2^4) \quad (1.7)$$

Ohm's law for heat transfer

$$\Delta T = \dot{Q}R \quad (1.8)$$

2 Conduction Through Fins

Formulas

$m = \sqrt{hp/kA_c}$, p is the perimeter and A_c is the cross-sectional area of the fin.

Very long fin:

$$\frac{T(x) - T_\infty}{T_b - T_\infty} = e^{-x\sqrt{hp/kA_c}} \quad (2.1)$$

$$\dot{Q}_{\text{long fin}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hpkA_c} (T_b - T_\infty) \quad (2.2)$$

Adiabatic fin tip:

$$\frac{T(x) - T_\infty}{T_b - T_\infty} = \frac{\cosh m(L - x)}{\cosh mL} \quad (2.3)$$

$$\dot{Q}_{\text{adiabatic tip}} = -kA_c \left. \frac{dT}{dx} \right|_{x=0} = \sqrt{hpkA_c} (T_b - T_\infty) \tanh mL \quad (2.4)$$

Specified temperature at fin tip:

$$\frac{T(x) - T_\infty}{T_b - T_\infty} = \frac{[(T_L - T_\infty) / (T_b - T_\infty)] \sinh mx + \sinh m(L - x)}{\sinh mL} \quad (2.5)$$

$$\dot{Q}_{\text{specified temp.}} = \sqrt{hpkA_c} (T_b - T_\infty) \frac{\cosh mL - [(T_L - T_\infty) / (T_b - T_\infty)]}{\sinh mL} \quad (2.6)$$

Convection from fin tip:

$$\frac{T(x) - T_\infty}{T_b - T_\infty} = \frac{\cosh m(L - x) + (h/mk) \sinh m(L - x)}{\cosh mL + (h/mk) \sinh mL} \quad (2.7)$$

$$\dot{Q}_{\text{convection}} = \sqrt{hpkA_c} (T_b - T_\infty) \frac{\sinh mL + (h/mk) \cosh mL}{\cosh mL + (h/mk) \sinh mL} \quad (2.8)$$

Fin efficiency:

$$\eta_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{fin,max}}} = \frac{\text{actual heat transfer from the fin}}{\text{heat transfer if the entire fin were at base temperature}} \quad (2.9)$$

Fin effectiveness:

$$\varepsilon_{\text{fin}} = \frac{\dot{Q}_{\text{fin}}}{\dot{Q}_{\text{no fin}}} \quad (2.10)$$

$$\varepsilon_{\text{fin}} = \frac{A_{\text{fin}}}{A_{\text{base}}} \eta_{\text{fin}} \quad (2.11)$$

Fin efficiency table and graphs

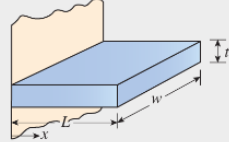
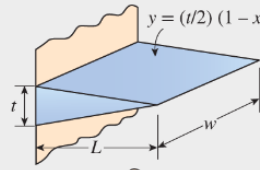
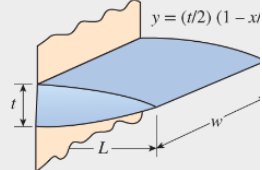
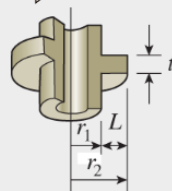
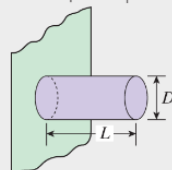
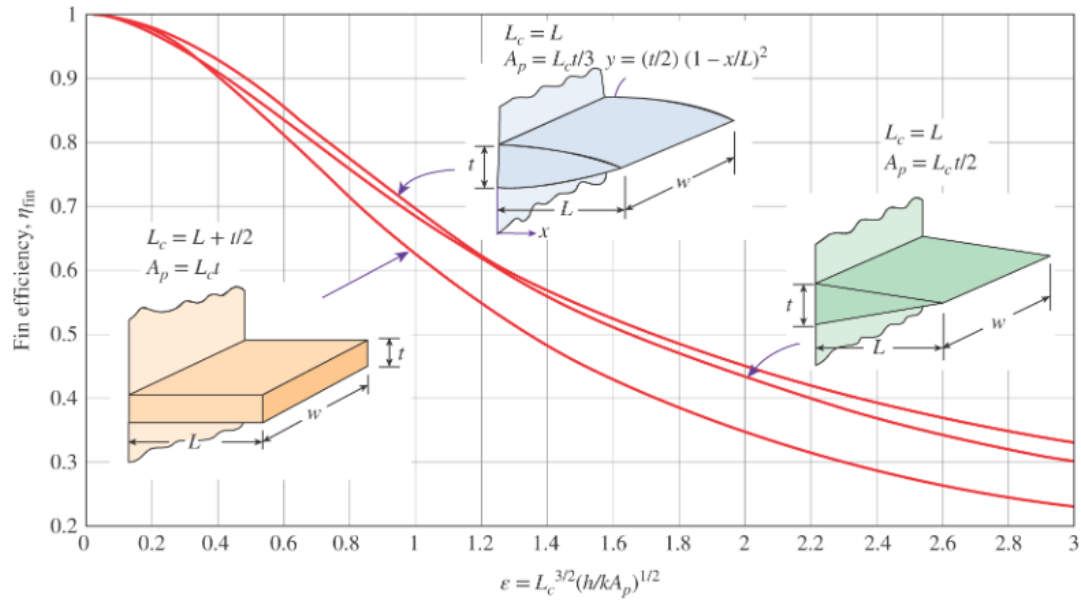
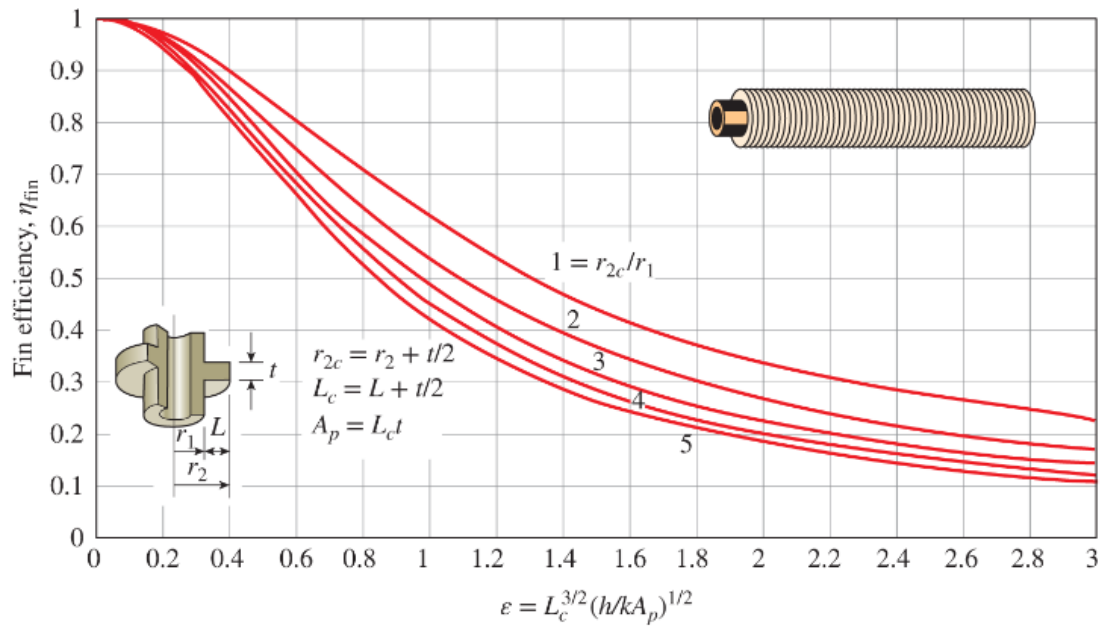
Straight rectangular fins $m = \sqrt{2h/kt}$ $L_c = L + t/2$ $A_{fin} = 2wL_c$	$\eta_{fin} = \frac{\tanh mL_c}{mL_c}$	
Straight triangular fins $m = \sqrt{2h/kt}$ $A_{fin} = 2w\sqrt{L^2 + (t/2)^2}$	$\eta_{fin} = \frac{1}{mL} \frac{I_1(2mL)}{I_0(2mL)}$	
Straight parabolic fins $m = \sqrt{2h/kt}$ $A_{fin} = wL[C_1 + (L/t)\ln(t/L + C_1)]$ $C_1 = \sqrt{1 + (t/L)^2}$	$\eta_{fin} = \frac{2}{1 + \sqrt{(2mL)^2 + 1}}$	
Circular fins of rectangular profile $m = \sqrt{2h/kt}$ $r_{2c} = r_2 + t/2$ $A_{fin} = 2\pi(r_{2c}^2 - r_1^2)$	$\eta_{fin} = C_2 \frac{K_1(mr_1)I_1(mr_{2c}) - I_1(mr_1)K_1(mr_{2c})}{I_0(mr_1)K_1(mr_{2c}) + K_0(mr_1)I_1(mr_{2c})}$ $C_2 = \frac{2r_1/m}{r_{2c}^2 - r_1^2}$	
Pin fins of rectangular profile $m = \sqrt{4h/kD}$ $L_c = L + D/4$ $A_{fin} = \pi DL_c$	$\eta_{fin} = \frac{\tanh mL_c}{mL_c}$	

Table 1: Fin efficiency for common shapes



Graph 2.1: Fin efficiency of straight fins



Graph 2.2: Fin efficiency of circular fins

3 Forced External Convection

Formulas

Reynolds Number

$$\text{Re} = \frac{vL_c}{\nu} \quad (3.1)$$

Nusselt Number

$$\text{Nu} = \frac{hL_c}{k} \quad (3.2)$$

Prandtl Number

$$\text{Pr} = \frac{\nu}{\alpha} = \frac{\mu c_p}{k} \quad (3.3)$$

The average Nusselt number for flow over a flat plate:

$$\text{Laminar: } \text{Nu} = \frac{hL}{k} = 0.664 \text{Re}_L^{0.5} \text{Pr}^{1/3}, \quad \text{Re} < 5 \times 10^5 \quad (3.4)$$

$$\text{Turbulent: } \text{Nu} = \frac{hL}{k} = 0.037 \text{Re}^{0.8} \text{Pr}^{1/3}, \quad 0.6 \leq \text{Pr} \leq 60, \quad 5 \times 10^5 \leq \text{Re} \leq 10^7 \quad (3.5)$$

$$\text{Combined: } \text{Nu} = \frac{hL}{k} = (0.037 \text{Re}^{0.8} - 871) \text{Pr}^{1/3}, \quad 0.6 \leq \text{Pr} \leq 60, \quad 5 \times 10^5 \leq \text{Re} \leq 10^7 \quad (3.6)$$

The average Nusselt numbers for crossflow over a *cylinder* and *sphere* are:

$$\text{Nu}_{\text{cyl}} = \frac{hD}{k} = 0.3 + \frac{0.62 \text{Re}^{1/2} \text{Pr}^{1/3}}{[1 + (0.4/\text{Pr})^{2/3}]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000} \right)^{5/8} \right]^{4/5}, \quad \text{Re Pr} > 0.2 \quad (3.7)$$

and

$$\text{Nu}_{\text{sph}} = \frac{hD}{k} = 2 + [0.4 \text{Re}^{1/2} + 0.06 \text{Re}^{2/3}] \text{Pr}^{0.4} \left(\frac{\mu_\infty}{\mu_s} \right)^{1/4}, \quad 3.5 \leq \text{Re} \leq 8 \times 10^4, 0.7 \leq \text{Pr} \leq 380 \quad (3.8)$$

The fluid properties are evaluated at the film temperature in the case of a cylinder, and at the free-stream temperature T_∞ (except for μ_s , which is evaluated at the surface temperature T_s) in the case of a sphere.

4 Forced Internal Convection

Formulas

Laminar hydrodynamic entry length:

$$L_{h, \text{ lam}} = 0.05 \text{ Re } D \quad (4.1)$$

Laminar thermal entry length:

$$L_{th, \text{ lam}} = 0.05 \text{ Re } \text{Pr} D = \text{Pr} L_{h, \text{ lam}} \quad (4.2)$$

Turbulent hydrodynamic and thermal entry length:

$$L_{h, \text{ turb}} = L_{th, \text{ turb}} = 10D \quad (4.3)$$

Fluid Temperature profile for constant surface heat flux $\dot{q}_s$:

$$T(x) = T_i + \frac{\dot{q}_s p}{m \dot{c}_p} x, \quad p = \text{perimeter} = \pi D \quad (4.4)$$

Surface Temperature profile for constant surface heat flux $\dot{q}_s$:

$$T_s(x) = T_i + \frac{\dot{q}_s p}{m \dot{c}_p} x + \frac{\dot{q}_s}{h} = T_i + \dot{q}_s \left(\frac{p h x + m \dot{c}_p}{h m \dot{c}_p} \right) \quad (4.5)$$

Fluid Temperature profile for constant surface temperature T_s :

$$T(x) = T_s - (T_s - T_i) \exp \left(\frac{-h p x}{m \dot{c}_p} \right), \quad p = \text{perimeter} = \pi D \quad (4.6)$$

Log mean temperature difference

$$\Delta T_{\text{lm}} = \frac{T_i - T_e}{\ln [(T_s - T_e)/(T_s - T_i)]} = \frac{\Delta T_e - \Delta T_i}{\ln (\Delta T_e / \Delta T_i)} \quad (4.7)$$

Total heat flux for constant surface temperature

$$\dot{Q} = h A_s \Delta T_{\text{lm}} \quad (4.8)$$

Nusselt number for fully developed laminar flow with constant heat flux

$$\text{Nu} = \frac{h D}{k} = 4.36 (= \text{constant}) \quad (4.9)$$

Nusselt number for fully developed laminar flow with constant surface temperature

$$\text{Nu} = \frac{h D}{k} = 3.66 (= \text{constant}) \quad (4.10)$$

Nusselt number for fully developed turbulent flow in smooth and rough tubes

$$\text{Nu} = 0.125 f \text{Re } \text{Pr}^{1/3} \quad (4.11)$$

Friction factor for smooth tubes

$$f = (0.79 \ln \text{Re} - 1.64)^{-2} \quad (4.12)$$

Nusselt number for turbulent flow in smooth tubes (more precise than (4.11))

$$\text{Nu} = 0.023 \text{Re}^{0.8} \text{Pr}^n \quad \text{with } n = 0.4 \text{ for heating and } n = 0.3 \text{ for cooling} \quad (4.13)$$

Nusselt number for fully developed laminar flow in an annulus with one surface isothermal and the other adiabatic

$$\text{Nu}_{i/o} = \frac{h_{i/o}(D_o - D_i)}{k} \quad (4.14)$$

D_i/D_o	Nu_i	Nu_o
0	–	3.66
0.05	17.46	4.06
0.10	11.56	4.11
0.25	7.37	4.23
0.50	5.74	4.43
1.00	4.86	4.86

Table 4.1: Nusselt number for annulus

5 Natural External Convection

Formulas

Grashof Number

$$\text{Gr} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu^2} \quad (5.1)$$

Raleigh Number

$$\text{Ra} = \text{Gr Pr} = \frac{g\beta(T_s - T_\infty)L_c^3}{\nu\alpha} \quad (5.2)$$

Nusselt number over surfaces

Geometry	L_c	Range of Ra	Nu
Vertical plate	L	$10^4 - 10^9$ $10^9 - 10^{13}$ Entire range	$\text{Nu} = 0.59\text{Ra}^{1/4}$ $\text{Nu} = 0.1\text{Ra}^{1/3}$ $\text{Nu} = \left\{0.825 + \frac{0.387\text{Ra}^{1/6}}{[1+(0.492/\text{Pr})^{9/16}]^{8/27}}\right\}^2$
Inclined plate	L		Use vertical plate equations for the upper surface of a cold plate and the lower surface of a hot plate. Replace g with $g \cos \theta$ for $0 < \theta < 60^\circ$.
Horizontal plate (surface area A_s and perimeter p) (a) Upper surface of a hot plate (or lower surface of a cold plate) (b) Lower surface of a hot plate (or upper surface of a cold plate)	A_f/p	$10^4 - 10^7$ $10^7 - 10^{11}$ $10^5 - 10^{11}$	(a) $\text{Nu} = 0.54\text{Ra}^{1/4}$ (a) $\text{Nu} = 0.15\text{Ra}^{1/3}$ (b) $\text{Nu} = 0.27\text{Ra}^{1/4}$
Vertical cylinder	L		A vertical cylinder can be treated as a vertical plate when $D \geq \frac{35L}{\text{Gr}^{1/4}}$.
Horizontal cylinder	D	$\text{Ra} \leq 10^{12}$	$\text{Nu} = \left\{0.6 + \frac{0.387\text{Ra}^{1/6}}{[1+(0.559/\text{Pr})^{9/16}]^{8/27}}\right\}^2$
Sphere	D	$\text{Ra} \leq 10^{11}$ ($\text{Pr} \geq 0.7$)	$\text{Nu} = 2 + \frac{0.589\text{Ra}^{1/4}}{[1+(0.469/\text{Pr})^{9/16}]^{4/9}}$

Table 5.1: Empirical correlations for the average Nusselt number for natural convection over surfaces

6 Natural Internal Convection

Formulas

Horizontal Rectangular Enclosures

$$\text{Nu} = 1 + 1.144 \left[1 - \frac{1708}{\text{Ra}} \right]^+ + \left[\frac{\text{Ra}^{1/3}}{18} - 1 \right]^+ \quad \text{Ra} < 10^8 \quad (6.1)$$

Inclined Rectangular Enclosures

$$\text{Nu} = 1 + 1.44 \left[1 - \frac{1708}{\text{Ra} \cos \theta} \right]^+ \left(1 - \frac{1708(\sin 1.8\theta)^{1.6}}{\text{Ra} \cos \theta} \right) + \left[\frac{(\text{Ra} \cos \theta)^{1/3}}{18} - 1 \right]^+ \quad (6.2)$$

$$\text{Ra} < 10^5 \quad 0 < \theta < 70^\circ \quad H/L \geq 12$$

Vertical Rectangular enclosures

$$\text{Nu} = 0.18 \left(\frac{\text{Pr}}{0.2 + \text{Pr}} \text{Ra} \right)^{0.29} \quad 1 < H/L < 2, \text{Ra} \frac{\text{Pr}}{0.2 + \text{Pr}} > 10^3 \quad (6.3)$$

$$\text{Nu} = 0.22 \left(\frac{\text{Pr}}{0.2 + \text{Pr}} \text{Ra} \right)^{0.28} \left(\frac{H}{L} \right)^{-1/4} \quad 2 < H/L < 10, \text{Ra} < 10^{10} \quad (6.4)$$

$$\text{Nu} = 0.42 \text{Ra}^{0.25} \left(\frac{H}{L} \right)^{-0.3} \quad 10 < H/L < 40, 1 < \text{Pr} < 2 \times 10^4, 10^4 < \text{Ra} < 10^7 \quad (6.5)$$

Concentric Cylinders ($L_c = (D_o - D_i)/2$)

$$\dot{Q} = \frac{2\pi k_{\text{eff}}}{\ln(D_o/D_i)} (T_i - T_o) \quad (6.6)$$

$$\frac{k_{\text{eff}}}{k} = 0.386 \left(\frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{cyl}} \text{Ra})^{1/4} \quad (6.7)$$

$$F_{\text{cyl}} = \frac{[\ln(D_o/D_i)]^4}{L_c^3 (D_i^{-3/5} + D_o^{-3/5})^5} \quad (6.8)$$

Concentric Spheres ($L_c = (D_o - D_i)/2$)

$$\dot{Q} = k_{\text{eff}} \frac{\pi D_i D_o}{L_c} (T_i - T_o) \quad (6.9)$$

$$\frac{k_{\text{eff}}}{k} = 0.74 \left(\frac{\text{Pr}}{0.861 + \text{Pr}} \right)^{1/4} (F_{\text{sph}} \text{Ra})^{1/4} \quad (6.10)$$

$$F_{\text{sph}} = \frac{L_c}{(D_i D_o)^4 (D_i^{-7/5} + D_o^{-7/5})^5} \quad (6.11)$$

7 Radiation

Formulas

Energy flux emitted by a black body

$$E_b = \sigma T^4 \quad (7.1)$$

Radiosity of grey surface with irradiation G

$$J = \varepsilon E_b + (1 - \varepsilon)G \quad (7.2)$$

Reciprocity Relation for view factors

$$A_i F_{i \rightarrow j} = A_j F_{j \rightarrow i} \quad (7.3)$$

Summation Rule for view factors

$$\sum_j F_{i \rightarrow j} = 1 \quad \left(\sum_i F_{i \rightarrow j} \neq 1 \right) \quad (7.4)$$

Superposition Rule for view factors

$$F_{(2,3) \rightarrow 1} = \frac{A_2 F_{2 \rightarrow 1} + A_3 F_{3 \rightarrow 1}}{A_2 + A_3} \quad (7.5)$$

Crossed string method for infinite surfaces

$$F_{i \rightarrow j} = \frac{\sum(\text{crossed strings}) - \sum(\text{uncrossed strings})}{2 \times \text{length of surface } i} \quad (7.6)$$

Surface Resistance

$$R_i = \frac{1 - \varepsilon_i}{A_i \varepsilon_i} \quad (R_i = \infty \text{ if } \varepsilon = 0) \quad (7.7)$$

Space Resistance

$$R_{ij} = \frac{1}{A_i F_{i \rightarrow j}} = \frac{1}{A_j F_{j \rightarrow i}} \quad (7.8)$$

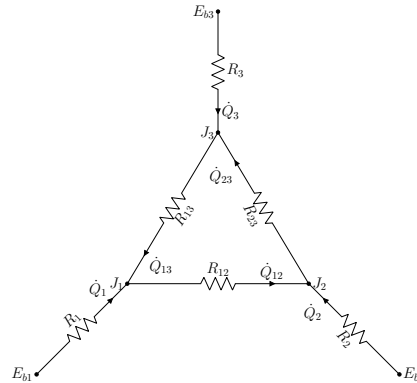
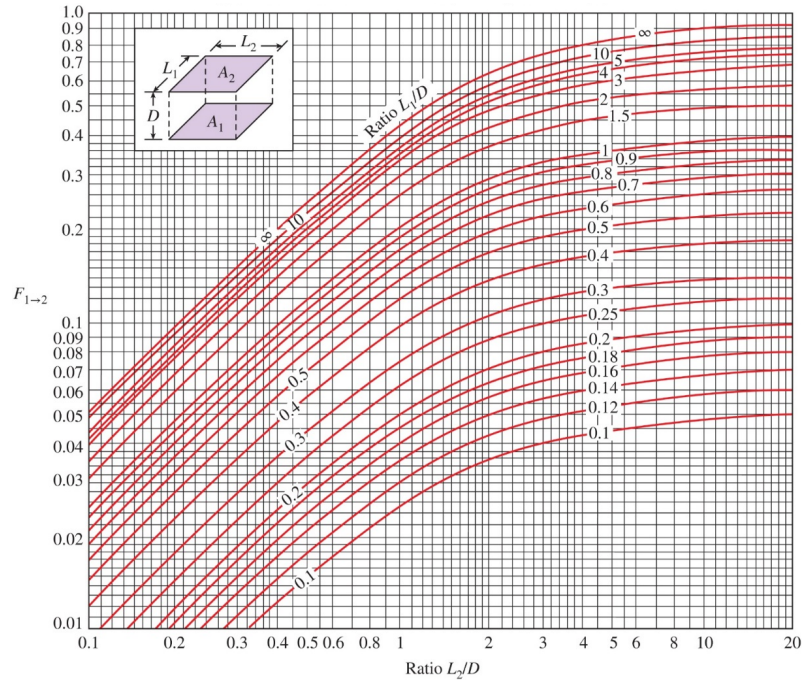
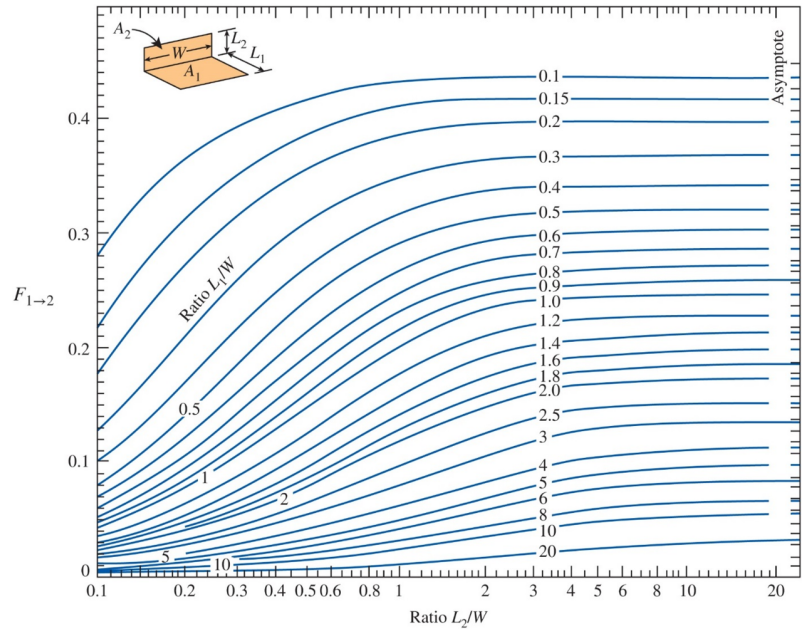


Figure 7.1: Thermal resistance network for three grey surfaces

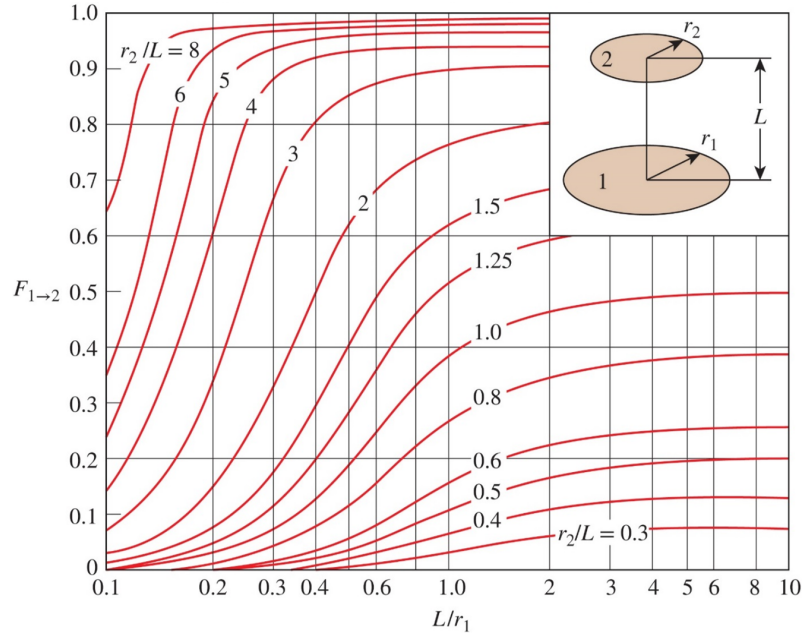
Graphs



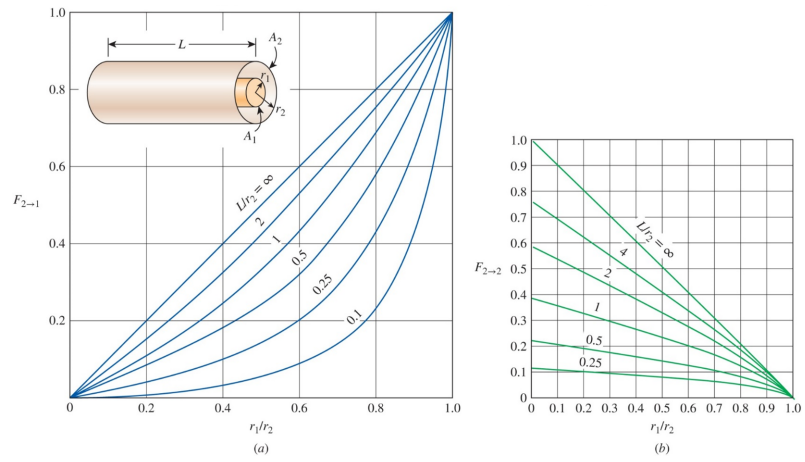
Graph 7.1: View factor between two identical parallel rectangles



Graph 7.2: View factor between two perpendicular plates with a common edge



Graph 7.3: View factor between two parallel coaxial disks



Graph 7.4: View factor between two concentric finite cylinders

8 Heat Exchangers

Formulas

Overall heat transfer coefficient U

$$\frac{1}{UA_s} = \frac{1}{U_i A_i} = \frac{1}{U_o A_o} = R_{\text{tot}} \quad (8.1)$$

For double-pipe exchangers

$$\frac{1}{UA_s} = R_{\text{tot}} = \frac{1}{h_i A_i} + \frac{\ln D_o/D_i}{2\pi k L} + \frac{1}{h_o A_o} \quad (8.2)$$

In the case of thin walls ($D_i = D_o$ and $A_i = A_o$)

$$U = U_i = U_o = \left(\frac{1}{h_i} + \frac{1}{h_o} \right)^{-1} \quad (8.3)$$

Heat capacity rate

$$C = \dot{m} c_p \quad (8.4)$$

Heat transfer in heat exchangers

$$\dot{Q} = UA_s \Delta T_m = C_h \Delta T_h = C_c \Delta T_c \quad (8.5)$$

For double-pipe heat exchangers (parallel flow and counterflow)

$$\dot{Q} = UA_s \Delta T_{\text{lm}} \quad (8.6)$$

with

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1/\Delta T_2)} \quad (8.7)$$

For other types of heat exchangers

$$\dot{Q} = UA_s F \Delta T_{\text{lm}} \quad (8.8)$$

Heat transfer effectiveness

$$\varepsilon = \frac{\dot{Q}}{\dot{Q}_{\text{max}}} = \frac{\text{actual heat transfer rate}}{\text{maximum possible heat transfer rate}} \quad (8.9)$$

Maximum possible heat transfer rate

$$\dot{Q}_{\text{max}} = C_{\text{min}}(T_{h,\text{in}} - T_{c,\text{in}}) \quad C_{\text{min}} = \min\{C_h, C_c\} \quad (8.10)$$

Number of transfer units

$$\text{NTU} = \frac{UA_s}{C_{\text{min}}} \quad (8.11)$$

capacity ratio

$$c = \frac{C_{\text{min}}}{C_{\text{max}}} \quad (8.12)$$

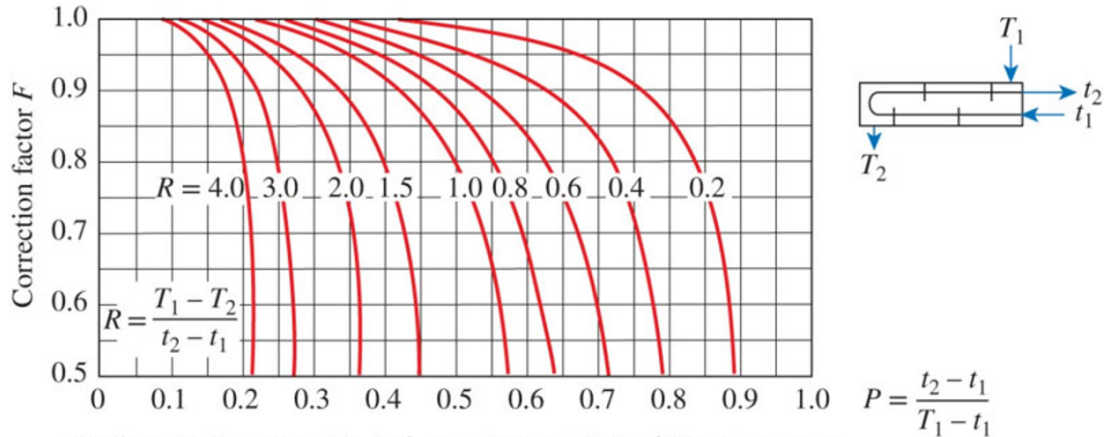
Tables and Graphs

Heat exchanger type	Effectiveness relation
1. Double pipe:	
<i>Parallel-flow</i>	$\varepsilon = \frac{1 - \exp[-NTU(1+c)]}{1+c}$
<i>Counterflow</i>	$\varepsilon = \frac{1 - \exp[-NTU(1+c)]}{1 - c \exp[-NTU(1-c)]} \quad (\text{for } c < 1)$
	$\varepsilon = \frac{NTU}{1+NTU} \quad (\text{for } c = 1)$
2. Shell-and-tube:	
<i>One shell pass 2, 4, ... tube passes</i>	$\varepsilon_1 = 2 \left\{ 1 + c + \sqrt{1 + c^2} \frac{1 + \exp[-NTU_1 \sqrt{1+c^2}]}{1 - \exp[-NTU_1 \sqrt{1+c^2}]} \right\}^{-1}$
<i>n shell passes 2n, 4n, ... tube passes</i>	$\varepsilon = \frac{1 - \exp[-NTU(1+c)]}{1+c}$
3. Crossflow (single-pass)	
<i>Both fluid unmixed</i>	$\varepsilon = 1 - \exp \left\{ \frac{NTU^{0.22}}{c} [\exp(-cNTU^{0.78}) - 1] \right\}$
<i>C_{max} mixed, C_{min} unmixed</i>	$\varepsilon = \frac{1}{c} (1 - \exp[-c(1 - \exp(-NTU))])$
<i>C_{min} mixed, C_{max} unmixed</i>	$\varepsilon = 1 - \exp \left\{ -\frac{1}{c} [1 - \exp(-cNTU)] \right\}$
4. All heat exchangers with c = 0	$\varepsilon = 1 - \exp(-NTU)$

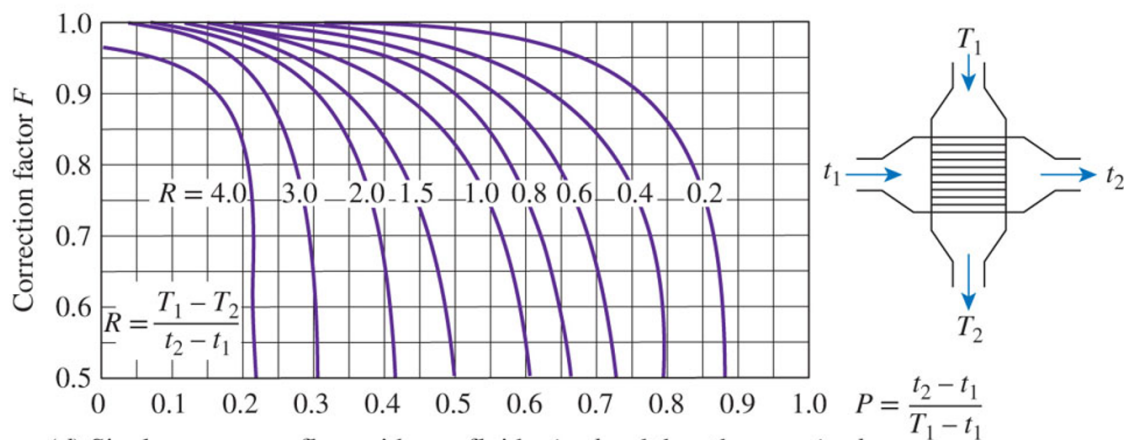
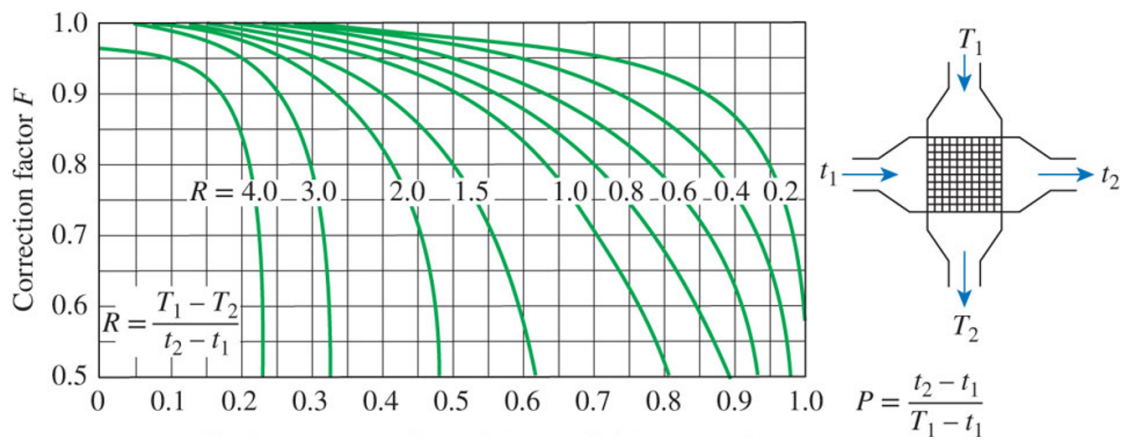
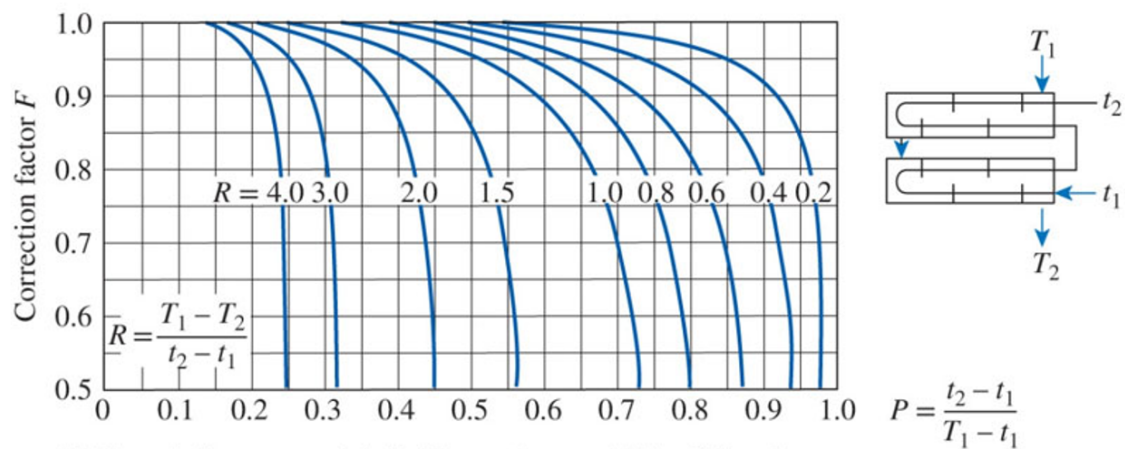
Table 8.1: Heat Exchanger Effectiveness - NTU Relations

Heat exchanger type	NTU relation
1. Double pipe:	
<i>Parallel-flow</i>	$NTU = -\frac{\ln[1-\varepsilon(1+c)]}{1+c}$
<i>Counterflow</i>	$NTU = \frac{1}{c-1} \ln\left(\frac{\varepsilon-1}{\varepsilon c-1}\right) \quad (\text{for } c < 1)$
	$NTU = \frac{\varepsilon}{1-\varepsilon} \quad (\text{for } c = 1)$
2. Shell and tube:	
<i>One shell pass 2, 4, ... tube passes</i>	$NTU = -\frac{1}{\sqrt{1+c^2}} \ln\left(\frac{2/\varepsilon_1-1-c-\sqrt{1+c^2}}{2/\varepsilon_1-1-c+\sqrt{1+c^2}}\right)$
<i>n shell passes 2n, 4n, ... tube passes</i>	$NTU = n(NTU)_1$
	To find effectiveness of the heat exchanger with one shell pass, use $\varepsilon_1 = \frac{F-1}{F-c}$ where $F = \left(\frac{\varepsilon c-1}{\varepsilon-1}\right)^{1/n}$
3. Crossflow (single-pass):	
<i>Both fluid unmixed</i>	(Use numerical method to invert the formula found in table 8.1)
$C_{\max}$ mixed, $C_{\max}$ unmixed	$NTU = -\ln\left[1 + \frac{\ln(1-\varepsilon c)}{c}\right]$
$C_{\min}$ mixed, $C_{\max}$ unmixed	$NTU = -\frac{\ln[c \ln(1-\varepsilon)+1]}{c}$
4. All heat exchangers with $c = 0$	$NTU = -\ln(1-\varepsilon)$

Table 8.2: Heat Exchanger NTU - Effectiveness Relations



Graph 8.1: Correction factors for 1 shell pass and 2n tube passes



9 Transient Conduction

criterion for lumped system:

$$\text{Bi} = \frac{hL_c}{k} \leq 0.1 \quad \text{with } L_c = \frac{\text{Volume}}{\text{surface area}} \quad (9.1)$$

undimensionalized temperature:

$$\theta(x, t) = \frac{T(x, t) - T_\infty}{T_i - T_\infty} \quad (9.2)$$

Temperature change for a lumped system:

$$\theta(t) = e^{-bt} \quad \text{with } b = \frac{hA_s}{\rho V c_p} \quad (9.3)$$

undimensionalized time (Fourier number):

$$\tau = \frac{\alpha t}{L^2} \quad \text{or} \quad \frac{\alpha t}{r^2} \quad (9.4)$$

One-term approximation for an infinite plane wall with thickness $2L$ and $x = 0$ in the center:

$$\theta_w(x, t) = A_1 e^{-\lambda_1^2 \tau} \cos(\lambda_1 x / L) \quad (9.5)$$

One-term approximation for an infinite cylinder with radius r_0 :

$$\theta_c(r, t) = A_1 e^{-\lambda_1^2 \tau} J_0(\lambda_1 r / r_0) \quad (9.6)$$

One-term approximation for a sphere with radius r_0 :

$$\theta_s(r, t) = A_1 e^{-\lambda_1^2 \tau} \frac{\sin(\lambda_1 r / r_0)}{\lambda_1 r / r_0} \quad (9.7)$$

Amount of heat transferred at time t :

$$\text{Plane wall:} \quad \left(\frac{Q}{Q_{\max}} \right)_w = 1 - \theta_{0,w} \frac{\sin \lambda_1}{\lambda_1} \quad (9.8)$$

$$\text{Cylinder:} \quad \left(\frac{Q}{Q_{\max}} \right)_c = 1 - 2\theta_{0,c} \frac{J_1(\lambda_1)}{\lambda_1} \quad (9.9)$$

$$\text{Sphere:} \quad \left(\frac{Q}{Q_{\max}} \right)_s = 1 - 3\theta_{0,s} \frac{\sin \lambda_1 - \lambda_1 \cos \lambda_1}{\lambda_1^3} \quad (9.10)$$

**Coefficients used in the one-term approximate solution of transient one-dimensional
heat conduction in plane walls, cylinders, and spheres**

($Bi = hL/k$ for a plane wall of thickness $2L$, or hr_o/k for a cylinder or sphere of radius r_o)

Bi	Plane Wall		Cylinder		Sphere	
	λ_1 (rad)	A_1	λ_1 (rad)	A_1	λ_1 (rad)	A_1
0.01	0.0998	1.0017	0.1412	1.0025	0.1730	1.0030
0.02	0.1410	1.0033	0.1995	1.0050	0.2445	1.0060
0.04	0.1987	1.0066	0.2814	1.0099	0.3450	1.0120
0.06	0.2425	1.0098	0.3438	1.0148	0.4217	1.0179
0.08	0.2791	1.0130	0.3960	1.0197	0.4860	1.0239
0.10	0.3111	1.0161	0.4417	1.0246	0.5423	1.0298
0.20	0.4328	1.0311	0.6170	1.0483	0.7593	1.0592
0.30	0.5218	1.0450	0.7465	1.0712	0.9208	1.0880
0.40	0.5932	1.0580	0.8516	1.0931	1.0528	1.1164
0.50	0.6533	1.0701	0.9408	1.1143	1.1656	1.1441
0.60	0.7051	1.0814	1.0184	1.1345	1.2644	1.1713
0.70	0.7506	1.0918	1.0873	1.1539	1.3525	1.1978
0.80	0.7910	1.1016	1.1490	1.1724	1.4320	1.2236
0.90	0.8274	1.1107	1.2048	1.1902	1.5044	1.2488
1.0	0.8603	1.1191	1.2558	1.2071	1.5708	1.2732
2.0	1.0769	1.1785	1.5995	1.3384	2.0288	1.4793
3.0	1.1925	1.2102	1.7887	1.4191	2.2889	1.6227
4.0	1.2646	1.2287	1.9081	1.4698	2.4556	1.7202
5.0	1.3138	1.2403	1.9898	1.5029	2.5704	1.7870
6.0	1.3496	1.2479	2.0490	1.5253	2.6537	1.8338
7.0	1.3766	1.2532	2.0937	1.5411	2.7165	1.8673
8.0	1.3978	1.2570	2.1286	1.5526	2.7654	1.8920
9.0	1.4149	1.2598	2.1566	1.5611	2.8044	1.9106
10	1.4289	1.2620	2.1795	1.5677	2.8363	1.9249
20	1.4961	1.2699	2.2880	1.5919	2.9857	1.9781
30	1.5202	1.2717	2.3261	1.5973	3.0372	1.9889
40	1.5325	1.2723	2.3455	1.5993	3.0632	1.9942
50	1.5400	1.2727	2.3572	1.6002	3.0788	1.9962
100	1.5552	1.2731	2.3809	1.6015	3.1102	1.9989
∞	1.5708	1.2732	2.4048	1.6021	3.1416	2.0000

Table 9.1: Coefficients for different Biot numbers and geometries.

Values for the zeroth order Bessel function of the first kind J_0

x	$J_0(x)$
0	1.000000
0.1	0.997502
0.2	0.990025
0.3	0.977626
0.4	0.960398
0.5	0.938470
0.6	0.912005
0.7	0.881201
0.8	0.846287
0.9	0.807524
1.0	0.765198
1.1	0.719622
1.2	0.671133
1.3	0.620086
1.4	0.566855
1.5	0.511828
1.6	0.455402
1.7	0.397985
1.8	0.339986
1.9	0.281819
2.0	0.223891
2.1	0.166607
2.2	0.110362
2.3	0.055540
2.4	0.002508

Table 9.2: Values of the Bessel function J_0